

# Spacing in Math Mode

Xi Ching Yuang

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## Introduction

Assume we have the next sets

$$S = \{z \in \mathbb{C} \mid |z| < 1\} \quad \text{and} \quad S_2 = \partial S$$

## Spaces

$$\begin{aligned} f(x) &= x^2 + 3x + 2 \\ f(x) &= x^2 + 3x + 2 \\ f(x) &= x^2 + 3x + 2 \\ f(x) &= x^2 + 3x + 2 \\ f(x) &= x^2 + 3x + 2 \\ f(x) &= x^2 + 3x + 2 \\ f(x) &= x^2 + 3x + 2 \\ f(x) &= x^2 + 3x + 2 \end{aligned}$$

## Operators Spacing

$$\begin{aligned} 3ax+4by&=5cz \\ 3ax\j4by&+5cz \end{aligned}$$

## User-defined binary and relational operators

$$\begin{aligned} 34x^2a \# 13bc \\ 34x^2a \# 13bc \end{aligned}$$

## Reference Guide

| Description of Spacing Commands        |                                              |
|----------------------------------------|----------------------------------------------|
| LaTeX Code                             | Description                                  |
| <code>\</code>                         | space equal to the current font size(=18 mu) |
| <code>\,</code>                        | $\frac{3}{18}$ (=3 mu)                       |
| <code>\:</code>                        | $\frac{4}{18}$ (=4 mu)                       |
| <code>\;</code>                        | $\frac{5}{18}$ (=5 mu)                       |
| <code>\;</code>                        | $-\frac{3}{18}$ (=-3 mu)                     |
| <code>\ (space after backslash)</code> | equivalent of space in normal text           |
| <code>\quad</code>                     | twice of (=-36 mu)                           |